

Air University



Object Oriented Programming (Lab)

Year 2025

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Academic Year: 2024 – 2028

Department of Cyber Security

USAMA ARSHAD | 247141

BS – Cyber Security – Fall – 24 (A)

Date of Submission: April 09, 2025

Assignment No. 02**Question 01****Code**

Row (1 - 26)

```
1  #include <iostream>
2  #include <cstring>
3  using namespace std;
4
5  class Student
6  {
7  private:
8      string std_name;
9      int std_age;
10     float std_marks;
11     friend void Display(const Student& s);
12
13 public:
14     Student() // default constructor
15     {
16         std_name = "";
17         std_age = 0;
18         std_marks = 0.0;
19     }
20     Student(string name, int age, float marks) // parameterized constructor
21     {
22         std_name = name;
23         std_age = age;
24         std_marks = marks;
25     }
26 };
```

Row (27 - 50)

```
27 void Display(const Student& s)
28 {
29     cout << "\nStudent Details: " << endl;
30     cout << "Name: " << s.std_name << endl;
31     cout << "Age: " << s.std_age << endl;
32     cout << "Marks: " << s.std_marks << "%" << endl;
33 }
34 int main()
35 {
36     string name;
37     int age;
38     float marks;
39
40     cout << "\nEnter Student Name: ";
41     getline(cin, name);
42
43     cout << "Enter Student Age: ";
44     cin >> age;
45     while (age < 0)
46     {
47         cout << "Enter Valid Age (Greater Than 0): ";
48         cin >> age;
49     }
50
```

Row (51 – 64)

```
51     cout << "Enter Student Marks (%): ";
52     cin >> marks;
53     while (marks < 0 || marks > 100)
54     {
55         cout << "Enter Valid Marks % (Between 0 and 100): ";
56         cin >> marks;
57     }
58
59     Student s1(name, age, marks);
60     Display(s1);
61
62     cout << endl;
63     return 0;
64 }
```

Output:

```
Enter Student Name: Usama Arshad
Enter Student Age: 0
Enter Valid Age (Greater Than 0): 20
Enter Student Marks (%): -1
Enter Valid Marks % (Between 0 and 100): 110
Enter Valid Marks % (Between 0 and 100): 87

Student Details:
Name: Usama Arshad
Age: 20
Marks: 87%
```

Question 02**Code**

Row (1 - 26)

```
1  #include <iostream>
2  #include <cstring>
3  using namespace std;
4
5  class Student
6  {
7  private:
8      string std_name;
9      int std_roll_no;
10     float std_marks;
11     friend void stdReport(const Student &s);
12
13 public:
14     Student() // default constructor
15     {
16         std_name = "";
17         std_roll_no = 0;
18         std_marks = 0.0;
19     }
20     Student(string name, int roll_no, float marks) // parameterized constructor
21     {
22         std_name = name;
23         std_roll_no = roll_no;
24         std_marks = marks;
25     }
26 };
```

Row (27 – 50)

```
27  void stdReport(const Student &s)
28  {
29      cout << "\nStudent Report: " << endl;
30      cout << "Name: " << s.std_name << endl;
31      cout << "Roll No: " << s.std_roll_no << endl;
32      cout << "Marks: " << s.std_marks << "%" << endl;
33  }
34  int main()
35  {
36      string name;
37      int roll_no;
38      float marks;
39
40      cout << "\nEnter Student Name: ";
41      getline(cin, name);
42
43      cout << "Enter Student's Roll No: ";
44      cin >> roll_no;
45      while (roll_no <= 0)
46      {
47          cout << "Enter Valid Roll No (Greater than 0): ";
48          cin >> roll_no;
49      }
50
```

Row (51 – 64)

```
51      cout << "Enter Student's Marks %: ";
52      cin >> marks;
53      while (marks < 0 || marks > 100)
54      {
55          cout << "Enter Valid Marks % (Between 0 and 100): ";
56          cin >> marks;
57      }
58
59      Student s1(name, roll_no, marks);
60      stdReport(s1);
61
62      cout << endl;
63      return 0;
64  }
```

Output:

```
Enter Student Name: Usama Arshad
Enter Student's Roll No: 0
Enter Valid Roll No (Greater than 0): 247141
Enter Student's Marks %: -1
Enter Valid Marks % (Between 0 and 100): 120
Enter Valid Marks % (Between 0 and 100): 85

Student Report:
Name: Usama Arshad
Roll No: 247141
Marks: 85%
```

THE END